



深度学习中的高效计算方法

Efficient Computing of Deep Neural Networks

Lecture 01: Introduction

Bei Yu

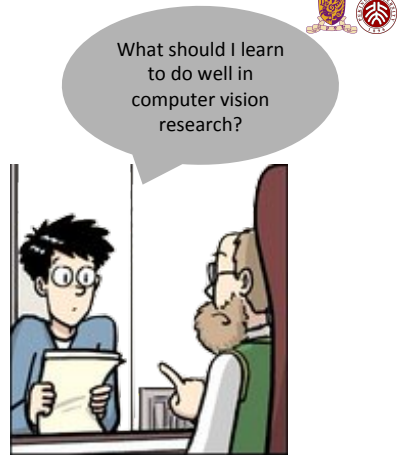
CSE Department, CUHK

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(Latest update: July 4, 2026)

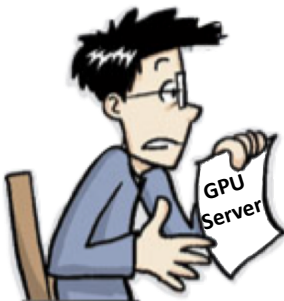
2025 Summer





DEEP LEARNING

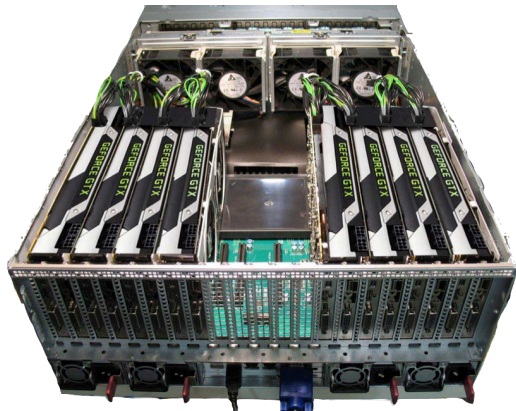






State of the art recognition methods

- Very Expensive
 - Memory
 - Computation
 - Power



HGX A100 4-GPU

Hyperplane 4-A100



- ✓ 4 NVIDIA® A100 SXM4 GPUs
- ✓ NVLink™ GPU connectivity
- ✓ 2 rack unit form factor (2U)
- ✓ 2x AMD EPYC processors with up to 64 cores
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Refer:

<https://shop.lambdalabs.com/deep-learning/servers/hyperplane/customize>



What We Focus on?



What you expect to Learn?

Introduction

Model level (Mo)

Mo1 Pruning

Mo2 Decomposition

Mo3 Quantization

Mo4 BNN

Mo5 KD

Mo6 NAS

Implementation level (Im)

Im1 GEMM

Im2 Direct Conv

Im3 Winograd

Im4 Sparse Conv

Im5 CUDA

Im6 TVM

① CNN Architecture Overview

② CNN Energy Efficiency

③ CNN on Embedded Platform

1 CNN Architecture Overview

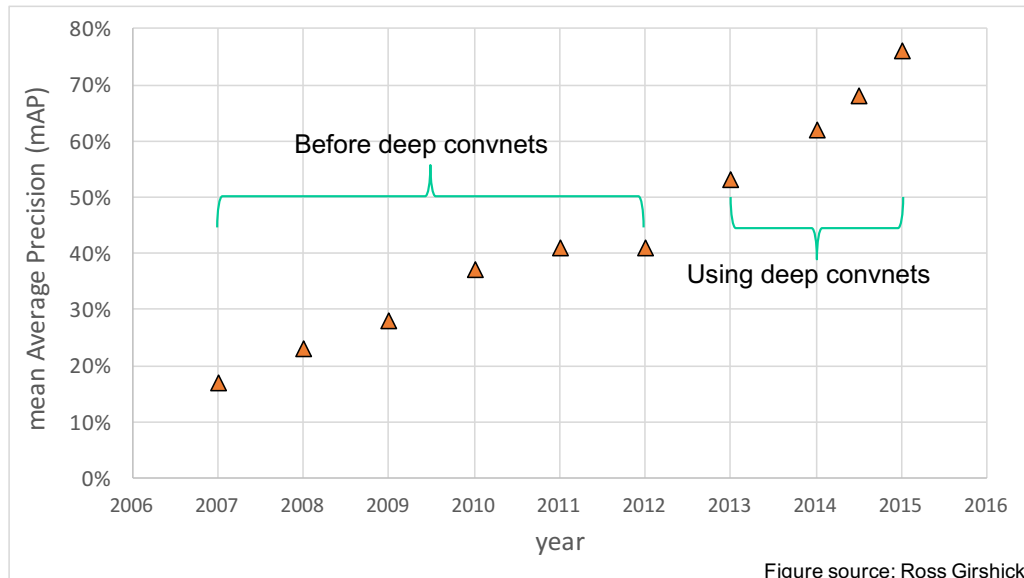
2 CNN Energy Efficiency

3 CNN on Embedded Platform

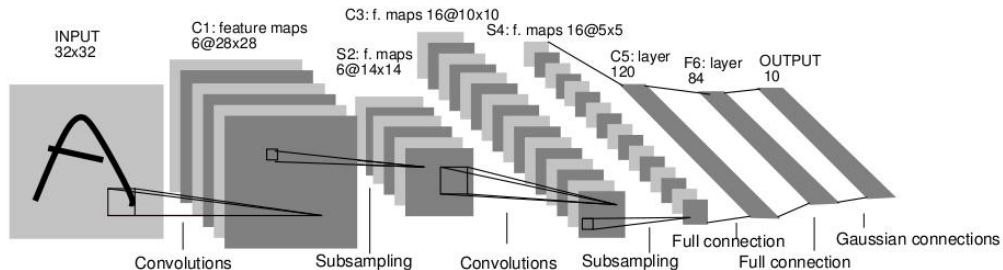
What happened to Object Detection



Object Detection: PASCAL VOC mean Average Precision (mAP)

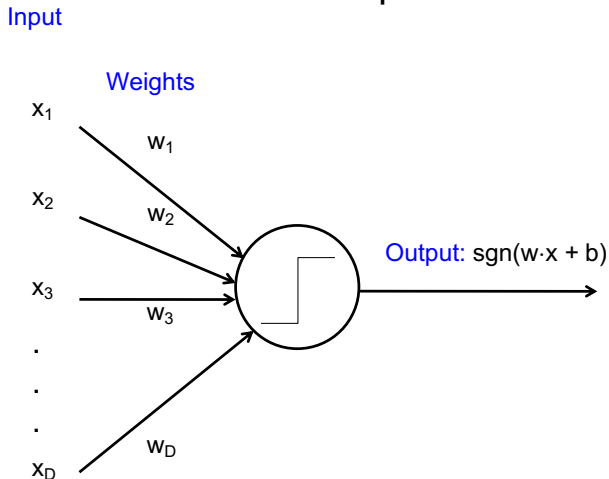


LeNet 5

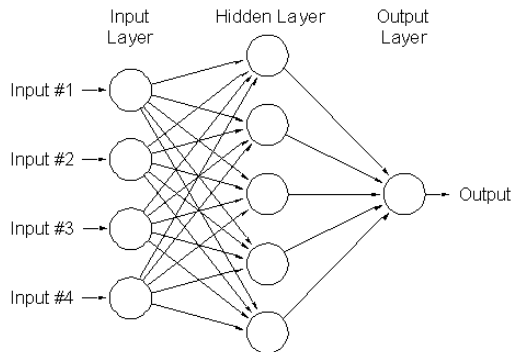


Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, [Gradient-based learning applied to document recognition](#), Proc. IEEE 86(11): 2278–2324, 1998.

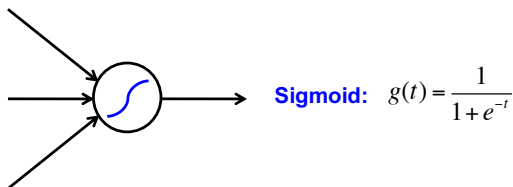
The Perceptron



Rosenblatt, Frank (1958), The Perceptron: A Probabilistic Model for Information Storage and Organization in the Brain, Cornell Aeronautical Laboratory, Psychological Review, v65, No. 6, pp. 386–408.

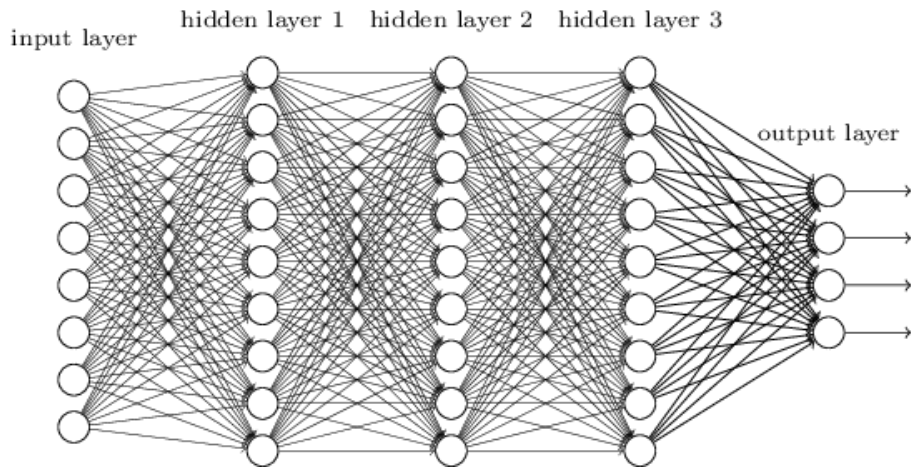


- Can learn nonlinear functions provided each perceptron has a differentiable nonlinearity



What is the value range of sigmoid activation?

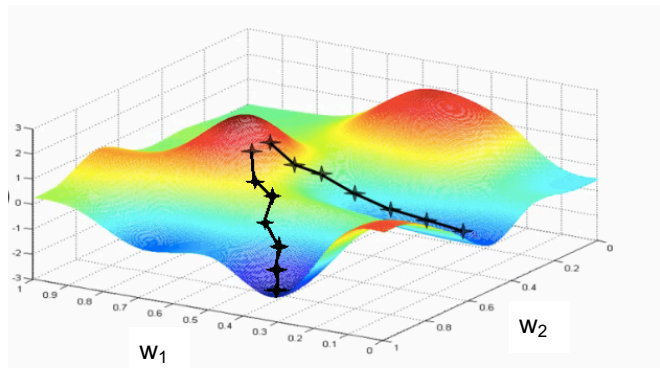
- $[-1, 1]$
- $[-\infty, +\infty]$
- $[0, 1]$
- $[0, +\infty]$



- Find network weights to minimize the *training error* between true and estimated labels of training examples, e.g.:

$$E(\mathbf{w}) = \sum_{i=1}^N (y_i - f_{\mathbf{w}}(\mathbf{x}_i))^2$$

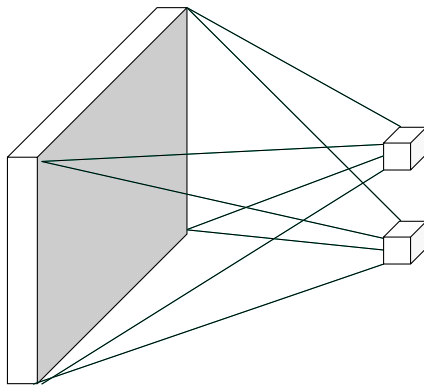
- Update weights by **gradient descent**: $\mathbf{w} \leftarrow \mathbf{w} - \alpha \frac{\partial E}{\partial \mathbf{w}}$



- Find network weights to minimize the *training error* between true and estimated labels of training examples, e.g.:

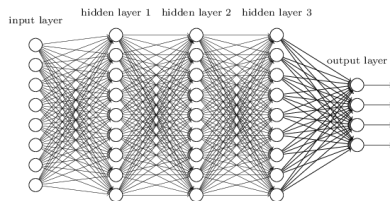
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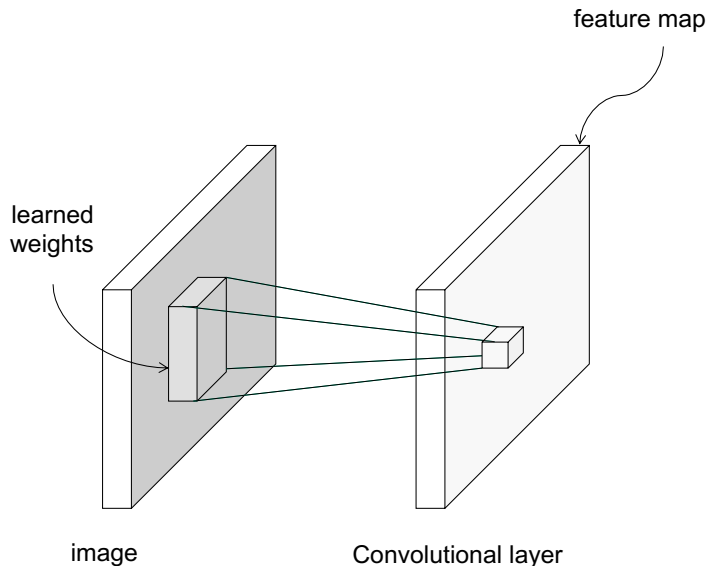
- Update weights by **gradient descent**: $\mathbf{w} \leftarrow \mathbf{w} - \alpha \frac{\partial E}{\partial \mathbf{w}}$
- Back-propagation**: gradients are computed in the direction from output to input layers and combined using chain rule
- Stochastic gradient descent**: compute the weight update w.r.t. one training example (or a small batch of examples) at a time, cycle through training examples in random order in multiple epochs



image

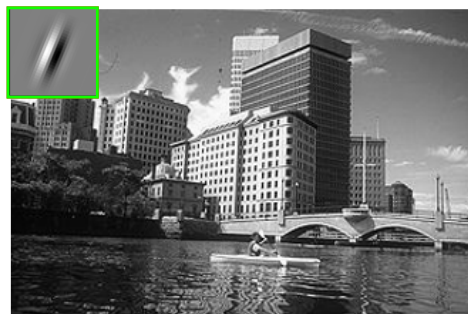
Fully connected layer



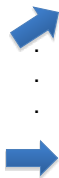


For a convolution kernel with kernel size 3, stride 1, what is the zero padding number to keep the output feature map size unchanged?

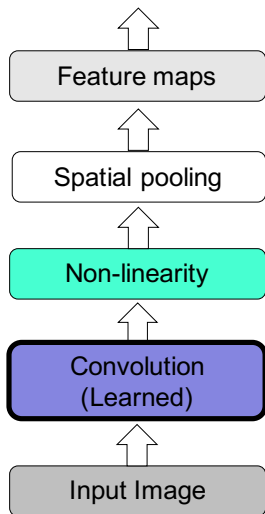
- A: 0
- B: 1
- C: 2
- D: 3



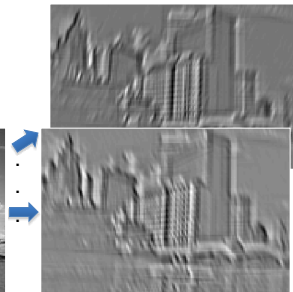
Input



Feature Map

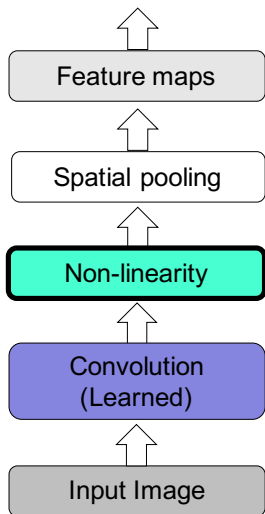


Input

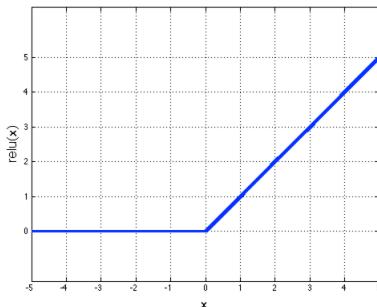


Feature Map

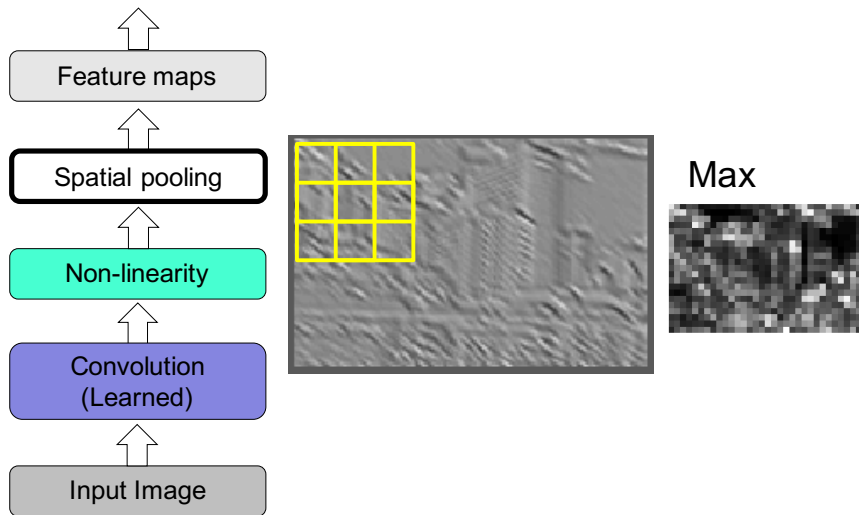
Source: R. Fergus, Y. LeCun



Rectified Linear Unit (ReLU)

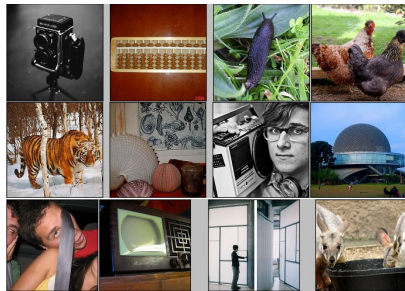


Source: R. Fergus, Y. LeCun



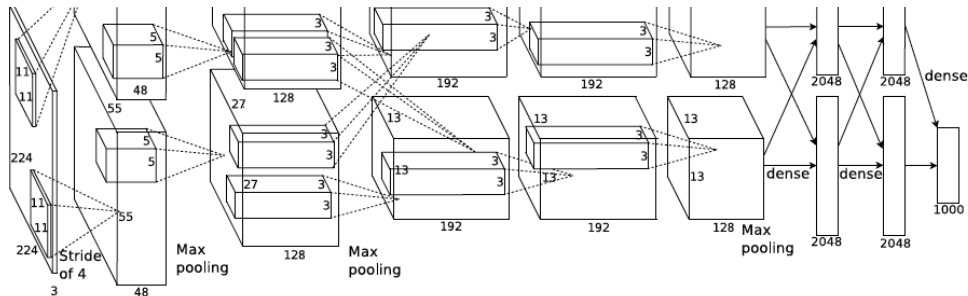
Source: R. Fergus, Y. LeCun

IMAGENET



- ~14 million labeled images, 20k classes
- Images gathered from Internet
- Human labels via Amazon MTurk
- ImageNet Large-Scale Visual Recognition Challenge (ILSVRC):
1.2 million training images, 1000 classes

www.image-net.org/challenges/LSVRC/



- Similar framework to LeNet but:
 - Max pooling, ReLU nonlinearity
 - More data and bigger model (7 hidden layers, 650K units, 60M params)
 - GPU implementation (50x speedup over CPU)
 - Trained on two GPUs for a week
 - Dropout regularization

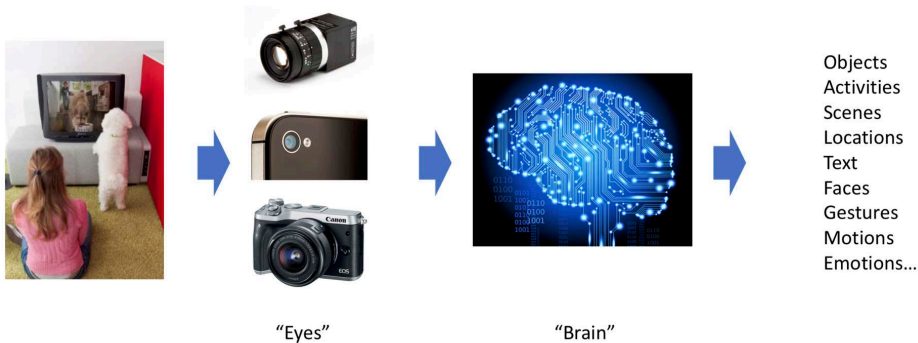
A. Krizhevsky, I. Sutskever, and G. Hinton, [ImageNet Classification with Deep Convolutional Neural Networks](#), NIPS 2012

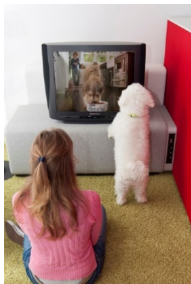
1 CNN Architecture Overview

2 CNN Energy Efficiency

3 CNN on Embedded Platform

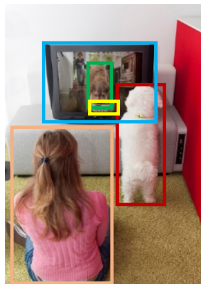
- Humans use their **eyes** and their **brains** to visually sense the world.
- Computers use their **cameras** and **computation** to visually sense the world





Classification

Image



Detection

Region



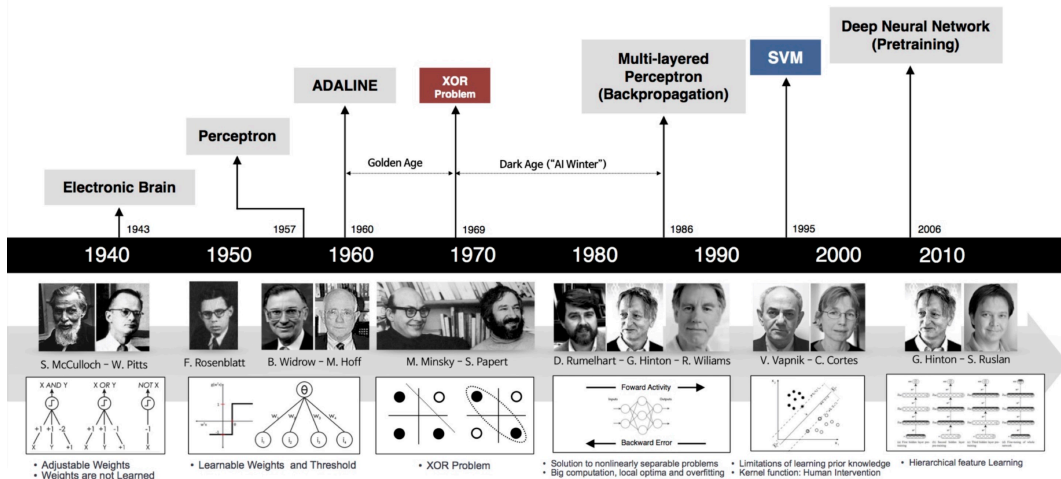
Segmentation

Pixel



Sequence

Video

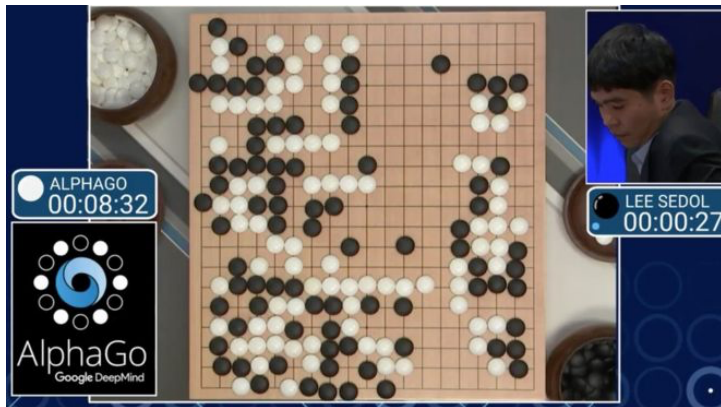


- The rises of SVM, Random forest
- No theory to play
- Lack of training data
- Benchmark is insensitive
- Difficulties in optimization
- Hard to reproduce results

Curse

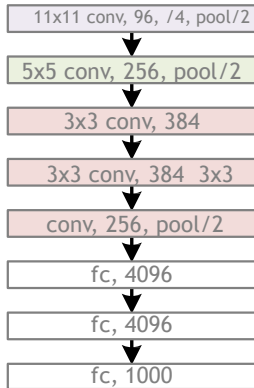
“Deep neural networks are no good and could never be trained.”

- A fast learning algorithm for deep belief nets. [Hinton et.al 1996]
- Data + Computing + Industry Competition
- NVidia's GPU, Google Brain (16,000 CPUs)
- **Speech**: Microsoft [2010], Google [2011], IBM
- **Image**: AlexNet, 8 layers [Krizhevsky et.al 2012] (26.2% -> 15.3%)



Revolution of Depth

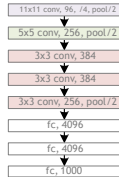
AlexNet, 8
layers
(ILSVRC 2012)



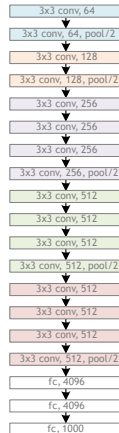
Slide Credit: He et al. (MSRA)

Revolution of Depth

AlexNet, 8
layers
(ILSVRC 2012)



VGG, 19
layers
(ILSVRC 2014)



GoogleNet, 22
layers
(ILSVRC 2014)



Slide Credit: He et al. (MSRA)

Revolution of Depth

AlexNet, 8
layers
(ILSVRC 2012)



VGG, 19
layers
(ILSVRC 2014)



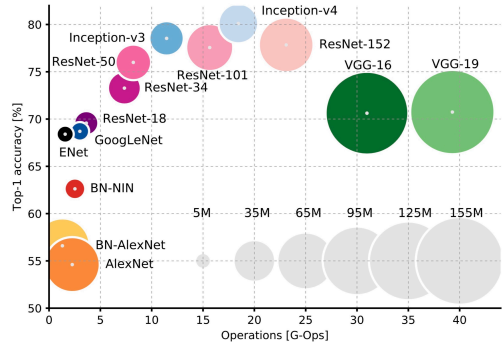
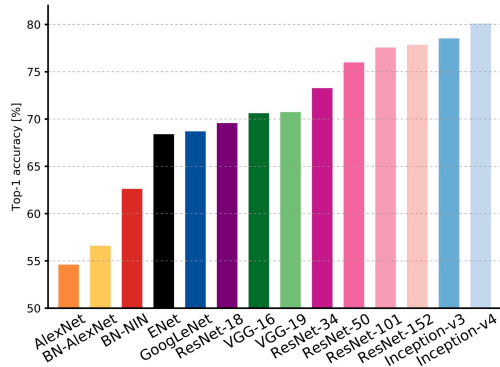
ResNet, 152
layers
(ILSVRC 2015)

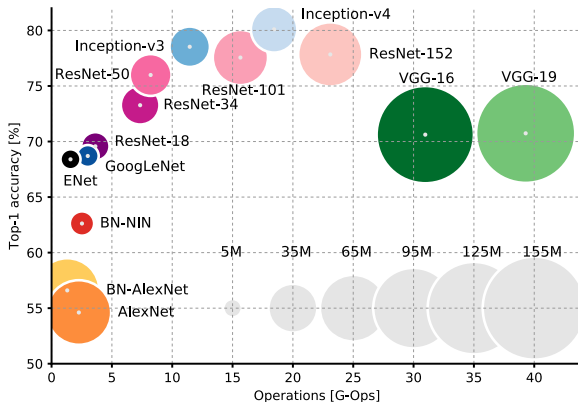


Slide Credit: He et al. (MSRA)

- AlexNet (Krizhevsky, Sutskever, and E. Hinton 2012) 233MB
- Network in Network (Lin, Chen, and Yan 2013) 29MB
- VGG (Simonyan and Zisserman 2015) 549MB
- GoogleNet (Szegedy, Liu, et al. 2015) 51MB
- ResNet (He et al. 2016) 215MB
- Inception-ResNet (Szegedy, Vanhoucke, et al. 2016)
- DenseNet (Huang et al. 2017)
- Xception (Chollet 2017)
- MobileNetV2 (Sandler et al. 2018)
- ShuffleNet (Zhang et al. 2018)

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- Inception-ResNet (Szegedy, Vanhoucke, et al. 2016) 23MB
- DenseNet (Huang et al. 2017) 80MB
- Xception (Chollet 2017) 22MB
- MobileNetV2 (Sandler et al. 2018) 14MB
- ShuffleNet (Zhang et al. 2018) 22MB





Why AlexNet is large in size, but small in operations?

- A: Special FC layers
- B: Special Conv layers
- C: More channels
- D: Some redundant operators

Autonomous drive

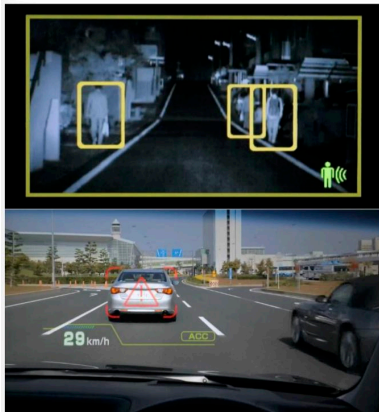
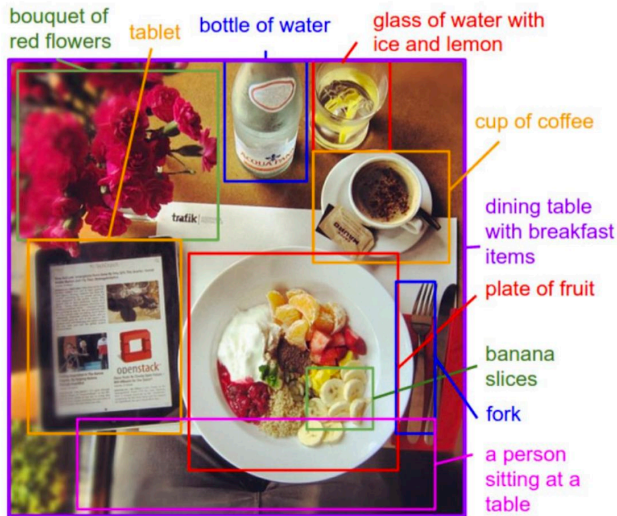
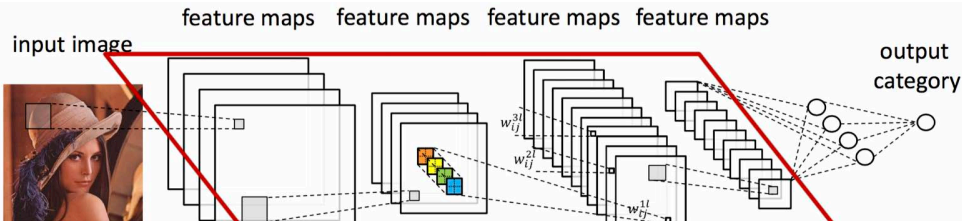


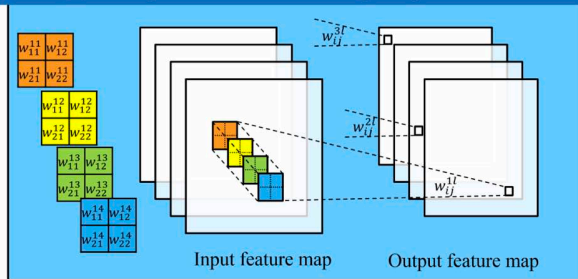
Image recognition



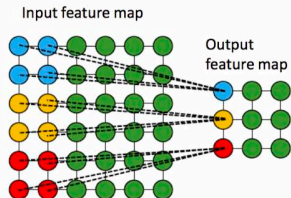


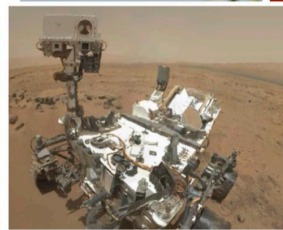
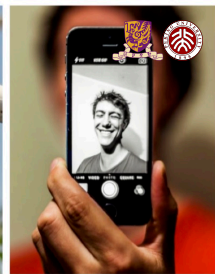
Convolutional layers account for over 90% computation

- [1] A. Krizhevsky, etc. Imagenet classification with deep convolutional neural networks. NIPS 2012.
- [2] J. Cong and B. Xiao. Minimizing computation in convolutional neural networks. ICANN 2014



Max-pooling is optional





Embedded CV



Hisense core|photonics

[Start Video](#)

① CNN Architecture Overview

② CNN Energy Efficiency

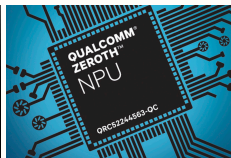
③ CNN on Embedded Platform

Convolution layer is one of the most expensive layers

- Computation pattern
- Emerging challenges

More and more end-point devices with limited memory

- Cameras
- Smartphone
- Autonomous driving

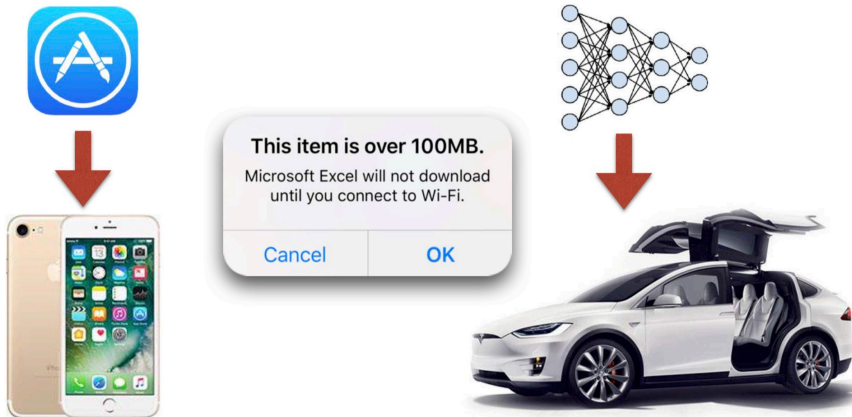


XILINX



An Intel Company

Hard to distribute large models through over-the-air update



2



AlphaGo: 1920 CPUs and 280 GPUs,
\$3000 electric bill per game



on mobile: **drains battery**
on data-center: **increases TCO**



3

³Song Han and William J. Dally (2018). "Bandwidth-efficient Deep Learning". In: *Proc. DAC*, 147:1–147:6.

Application Category

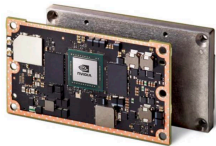


Both	Datacenter	Edge
Intel, Nvidia, IBM, Xilinx, HiSilicon, Google, Baidu, Alibaba Group, Cambricon, DeePhi, Bitmain, Wave Computing	AMD, Microsoft, Apple, Tencent Cloud, Aliyun, Baidu Cloud, HUAWEI Cloud, Fujitsu, Nokia, Facebook, HPE, Thinkforce, Cerebras, Graphcore, Groq, SambaNova Systems, Adapteva, PEZY	Qualcomm, Samsung, STMicroelectronics, NXP, MediaTek, Rockchip, Amazon_AWS, ARM, Synopsys, Imagination, CEVA, Cadence, VeriSilicon, Videantis, Horizon Robotics, Chipintelli, Unisound, AISpeech, Rokid, KnuEdge, Tenstorrent, ThinCI, Koniku, Knowm, Mythic, Kalray, BrainChip, Almotive, DeepScale, Leepmind, Krtkl, NovuMind, REM, TERADEEP, DEEP VISION, KAIST DNPu, Kneron, Esperanto Technologies, Gyrfalcon Technology, GreenWaves Technology, Lightelligence, Lightmatter, ThinkSilicon, Innogrit, Kortiq, Hailo, Tachyum

Source: <https://basicmi.github.io/Deep-Learning-Processor-List/>



CPU
(Raspberry Pi3)



GPU
(Jetson TX2)

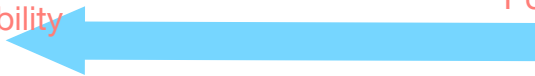


FPGA
(UltraZed)



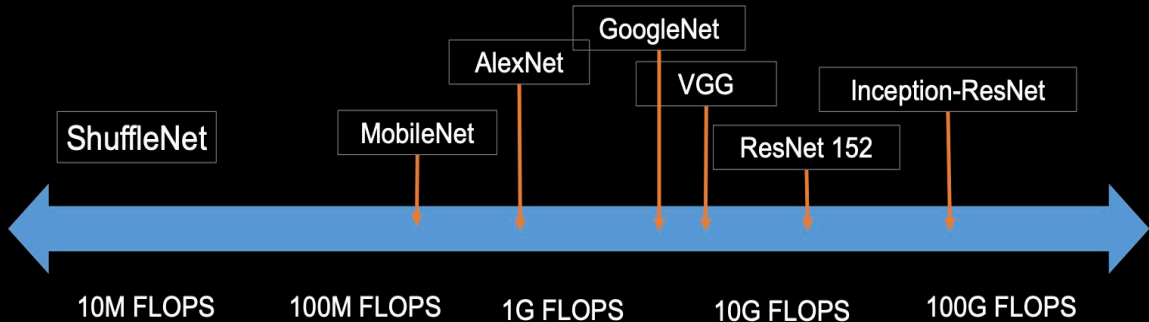
ASIC
(Movidius)

Flexibility



Power/Performance
Efficiency

Computing Spectrum



In-Datacenter Performance Analysis of a Tensor Processing Unit™

Norman P. Jouppi, Cliff Young, Nishant Patil, David Patterson, Gaurav Agrawal, Raminder Bajwa, Sarah Bates, Suresh Bhatia, Nan Boden, Al Borchers, Rick Boyle, Pierre-luc Cantin, Clifford Chao, Chris Clark, Jeremy Coriell, Mike Daley, Matt Dau, Jeffrey Dean, Ben Gelb, Tara Vazir Ghaemmaghami, Rajendra Gottipati, William Gulland, Robert Hagmann, C. Richard Ho, Doug Hogberg, John Hu, Robert Hundt, Dan Hurt, Julian Ibarz, Aaron Jaffey, Alek Jaworski, Alexander Kaplan, Harshit Khaitan, Daniel Killebrew, Andy Koch, Naveen Kumar, Steve Lacy, James Laudon, James Law, Diemthu Le, Chris Leary, Zhuyuan Liu, Kyle Lucke, Alan Lundin, Gordon MacKean, Adriana Maggiore, Maire Mahony, Kieran Miller, Rahul Nagarajan, Ravi Narayanaswami, Ray Ni, Kathy Nix, Thomas Norrie, Mark Omernick, Narayana Penukonda, Andy Phelps, Jonathan Ross, Matt Ross, Amir Salek, Emad Samadiani, Chris Severn, Gregory Sizikov, Matthew Snelham, Jed Souter, Dan Steinberg, Andy Swing, Mercedes Tan, Gregory Thorson, Bo Tian, Horia Toma, Erick Tuttle, Vijay Vasudevan, Richard Walter, Walter Wang, Eric Wilcox, and Doe Hyun Yoon

Google, Inc., Mountain View, CA USA

Email: {jouppi, cliffy, nishantpatil, davidpatterson}@google.com

To appear at the 44th International Symposium on Computer Architecture (ISCA), Toronto, Canada, June 26, 2017.

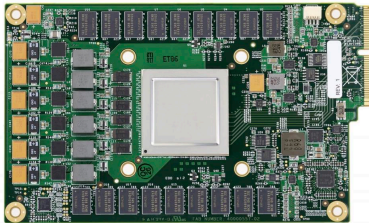


Figure 3. TPU Printed Circuit Board. It can be inserted in the slot for an SATA disk in a server, but the card uses PCIe Gen3 x16.

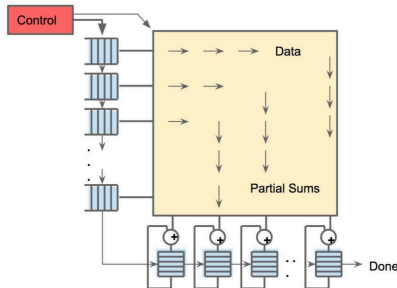
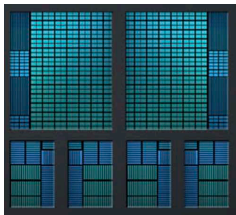


Figure 4. Systolic data flow of the Matrix Multiply Unit. Software has the illusion that each 256B input is read at once, and they instantly



- [Start Video](#)
- [Introduction Link 2](#)



(a) A11



(b) Kirin970



(c) Snapdragon845

Microsoft: FPGA Wins Versus Google TPUs For AI



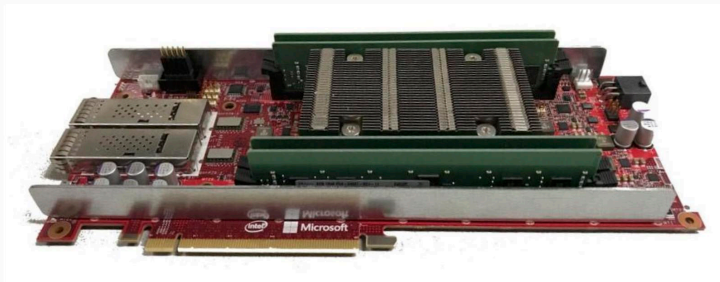
Moor Insights and Strategy Contributor ①

GUEST POST WRITTEN BY

Karl Freund

 Karl
Freund

Karl Freund is Sr. Analyst, Machine Learning and HPC, Moor Insights & Strategy



The Microsoft Brainwave mezzanine card extends each server with an Intel Altera Stratix 10 FPGA accelerator, synthesized to act as a "Soft DNN Processing Unit," or DPU, and a fabric interconnect that enables datacenter-scale persistent neural networks. MICROSOFT

FPGA云服务器

FPGA 云服务器 (FPGA Cloud Computing) 是基于FPGA (Field Programmable Gate Array) 现场可编程阵列的计算服务, 您只需单击几下即可在几分钟内轻松获取并部署您的FPGA计算实例。您可以在FPGA实例上编程, 为您的应用程序创建自定义硬件加速。我们为您提供可重编程的环境, 您可以在FPGA实例上多次编程, 而无需重新设计硬件, 让您能更加专注于业务发展。

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INTEL FGAS POWER ACCELERATION-AS-A-SERVICE FOR ALIBABA CLOUD

Intel today announced that Intel® field programmable gate arrays (FPGAs) are now powering the Acceleration-as-a-Service of Alibaba Cloud*, the cloud computing arm of Alibaba Group. The acceleration service, which can be launched from the Alibaba Cloud website, enables customers to develop and deploy accelerator solutions in the cloud for Artificial Intelligence inference, video streaming analytics, database acceleration and other fields where intense computing is required.



Xilinx Selected by Alibaba Cloud for Next-Gen FPGA Cloud Acceleration

Xilinx FPGAs are accelerating machine learning and other critical compute workloads for one of the world's largest cloud providers

Oct 12, 2017

HANGZHOU, China, Oct. 12, 2017 /PRNewswire/ -- Xilinx, Inc. (NASDAQ: XLNX) today announced at the Computing Conference that Alibaba Cloud, the cloud computing arm of Alibaba Group, has chosen Xilinx for next generation FPGA acceleration in their public cloud. As the largest cloud provider in China, Alibaba Cloud offers high-performance, elastic computing power to over two million customers. Based on Xilinx® FPGAs, the new "F2" instances give Alibaba Cloud customers access to acceleration for data analytics, genomics, video processing, and machine learning workloads.

NVIDIA CEO Says "FGPA is Not the Right Answer" for Accelerating AI

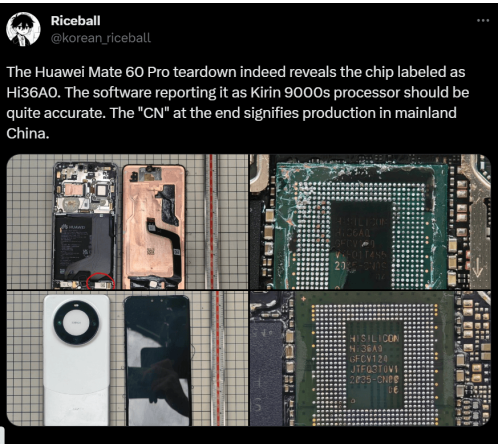


<https://medium.com/syncedreview/nvidia-ceo-says-fgpa-is-not-the-right-answer-for-accelerating-ai-83c810969edd>



	Xilinx ZCU102	Xilinx ZCU104	Huawei Atlas 200	nVIDIA Jetson TX2	Cambricon MLU 270
price	3K RMB	2K RMB	4K RMB	2.8K RMB	12K RMB
MobileNet-V1	1.14 ms	1.37 ms	1.8 ms	12.44 ms	1.85 ms
ResNet50	5.23 ms	6.81 ms	3.6 ms	24.70 ms	2.54 ms
Inception_v2	2.68 ms	3.35 ms	6.0 ms	10.81 ms	5.12 ms
Inception_v3	6.44 ms	8.53 ms	5.7 ms	32.53 ms	4.71 ms
Inception_v4	11.87 ms	17.06 ms	9.3 ms	44.37 ms	11.33 ms

⁴price is NOT accurate – reference purpose.



Mate 60 Pro's DNN solution is:

- A: CPU
- B: GPU
- C: FPGA
- D: ASIC